A Game-Theoretic Approach on Conflict Resolution of Autonomous Vehicles at Unsignalized Intersections

Xinghao Lu, Haiyan Zhao, Cheng Li, Bingzhao Gao, *Member, IEEE*, and Hong Chen, *Fellow, IEEE*

***Abstract*— In order to resolve the driving conflict and improve the safety and efficiency of autonomous vehicles at unsignalized intersections, a new decision-making method based on game theory with personalized driving preferences considered is proposed in this paper. In the decision-making method, the alterable game mode is constructed by combing the designed game entrying mechanism and replanning of game sequential order under different intersection conditions to ensure the adaptiveness and effectiveness of the decision-making algorithm. Besides, four payoff indicators and personalized payoff function are designed with the consideration of driving efficiency, safety and comfort requirement. The advantage of proposed method is that it not only reduces the complexity of game mode and improves the effectiveness at the same time, but also fully considers the personalized driving preferences to realize human-like driving and personalized decision while the existing methods usually neglect the difference among the drivers and its influence. Five different typical scenarios at the unsignalized intersection are given under co-simulation environment of Matlab and Prescan. The results show that the proposed decision-making method is able to resolve the driving conflict and ensure the safe passage of autonomous vehicles at unsignalized intersections, which verifies the effectiveness of the proposed method.**

***Index Terms*— Unsignalized intersection, driving conflict, autonomous vehicles.**

I. Introduction

THE development of intelligent vehicles is of great significance for improving traffic efficiency, reducing congestion and improving traffic safety, and has made great progress this year [1], [2], [3]. However, it is still severely limited in some complex traffic environments, such as unsignalized intersections especially [4], [5]. Due to the lack of guidance of traffic lights and complex traffic conditions, the passage sequence depends on the competition among drivers, which makes the unsignalized intersection prone to collision accidents, and the accident often causes traffic congestion, greatly reduces the road efficiency and driving safety [6], [7], [8], [9]. Therefore, it is an urgent need to construct an effective and safe method at unsignalized intersections for further avoiding accident risks and improving availability of autonomous driving [10], [11], [12].

The most traditional approach to the driving conflict of autonomous vehicles at unsignalized intersections is converting it into an optimization problem, under corresponding framework such as model predictive control, Bezier curve optimization method and so on [13], [14], [15]. Reference [16] proposed a centralized model predictive control method for the safe passage of autonomous vehicle at the unsignalized intersection with uncertainties of position measurements into considered. Reference [17] proposed a method that minimized the total energy consumption and driving time under optimal control framework for coordinating autonomous vehicles crossing an unsignalized intersection, and validate the effectiveness by numerical results. Reference [18] proposed a bezier curve optimization method to the obstacle avoidance problem of autonomous vehicle in intersection and solved it by the Lagrangian and Gradient-based method, which allows the safe passage in the condition that an obstacle appears. However, the optimization based method usually neglects the continuous interaction between vehicles and thus is not suitable when traffic scenario becomes complex.

Machine-learning approach have been suggested and well established in recent years with the development of artificial intelligence and computer science [19], [20], [21]. The main advantages lie in that low demand for prior knowledge, no need to manually formulate driving rules or driving systems and so on. Reference [22] proposed a method combining deep deterministic policy gradient algorithm and time-series Markov decision process algorithm to deal with the random states at complex intersection, which performs better in terms of convergence and generalizability than traditional deep deterministic policy gradient algorithm. Reference [23] proposed a reinforcement learning method to guarantee the driving safety of autonomous vehicles at intersection and addressed a belief update technique to adapt to perception errors. Reference [24] developed an integrated safety-enhanced framework combing reinforcement learning and model predictive control for autonomous vehicles to navigate unsignalized intersections, which has the advantage in preventing an

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Xinghao Lu, Haiyan Zhao, and Cheng Li are with the College of Communication Engineering, Jilin University, Changchun 130025, China (e-mail: luxh21@mails.jlu.edu.cn; zhao_hy@jlu.edu.cn; licheng21@mails.jlu.edu.cn).

Bingzhao Gao and Hong Chen are with the College of Electronic and Information Engineering, Tongji University, Shanghai 201804, China (e-mail: gaobz@jlu.edu.cn; chenh@jlu.edu.cn).

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uncontrolled and rapid switching behavior between algorithms. Reference [25] proposed a method centralized cooperation method for autonomous vehicles at the unsignalized intersection by using reinforcement learning, and simulation results show that the method can cut down the computation burden with the driving safety guaranteed. However, the machine learning based method universally exists the problem of difficulty for obtaining training data, long training time and poor interpretability.

Compared to the above mentioned approaches, the model based game theory has more advantage and superiority for the continuous interaction behavior decision making problem [26], [27], [28]. Besides, it is also widely used in other microscopic transportation fields such as signal control and lane changing and so on in recent years [29], [30], [31]. Reference [32] proposed a game-theoretic method to model the interactions of autonomous vehicles at unsignalized intersections, which is verified effective in reflecting the continuous interactions between vehicles. Reference [33] proposed the driving strategies for autonomous vehicles passing the unsignalised intersection safely under framework of game theory, and developed a tailored branch and bound algorithm to find the solution of games, and the simulation results show that it can be readily incorporated into real-time decision system. Reference [34] proposed a method under game-theoretic framework for modeling the interactive behavior of autonomous vehicles at the unsignalized intersection, and the simulation results show that the proposed method can effectively solve the driving conflicts at intersection.

To sum up, although some progress has been made in the conflict resolution at unsignalized intersections, there still exists the following problems or lacks need to be solved: 1) Most of the existing methods adopt the "one decision" method, ignoring the interaction between vehicles, which may lead to a low degree of anthropomorphic and rigid movement performance of vehicles. Moreover, the differences between vehicles are often ignored in the coordination algorithm, which is difficult to meet the personalized decision-making requirement. 2) Only two-vehicle game are involved and discussed in the most of game-theory based related studies, and the game among multiple vehicles is often simplified into a hierarchical continuous two-vehicle game, which is difficult to achieve unified group decision-making. Besides, the game mode is fixed and only depends on the position to determine the game vehicle and lacks the exit mechanism, which increases the unnecessary calculation burden. 3) The game sequential order is generally set as fixed pattern in the whole game process simply based on the clockwise direction or initial position, leading the sequential order cannot be adjusted according to the actual traffic situation and may cause unnecessary waiting or even congestion.

Aiming at the main exsiting problems mentioned above, a new decision-making method based on game theory at unsignalized intersections is proposed and the contributions of this paper are: 1) We propose a novel decision making framework based on non-cooperative dynamic game algorithm to solve the problem of human-like interaction between autonomous vehicles and safe allocation of the right of way at unsignalized intersections, and the driving demand is modeled in detail with the consideration of driving efficiency, safety and comfort requirement (taking driving style into integrated at the same time) to realize the human-like decision. 2) We propose the game entry/exit mechanism to judge the autonomous vehicle participating in the interactive game in real time and screen out the unnecessary vehicles at the same time, and establish the alterable game mode based on actual driving condition, so as to reduce the complexity of the game system and simplify the calculation of decision-making problems. 3) We propose the game sequential adjustment mechanism based on the time to reach the conflict position, which allows the sequential sequence to be adjusted at each stage of the game according to specific driving situation, and makes it easier for vehicles with higher risks to choose actions with greater benefits, which is conducive to ensuring the safe passage of the whole vehicles to some extent. 4) We demonstrate the effectiveness of the proposed method under co-simulation platform of Matlab and Prescan with different typical scenarios at unsignalized intersections. Based on the simulation results, we show that the proposed decision-making method is able to resolve the driving conflict and ensure the safe passage of connected autonomous vehicles at unsignalized intersections.

The remainder of this paper is organized as follows. The problem formulation and game theoretic decision-making framework for connected autonomous vehicles at unsignalized intersection are introduced in Section II, where Section II-A designs the game entrying mechanism and constructs the alterable game mode. Section II-B designs the four payoff indicators with the consideration of different driving requirements, and then constructs personalized payoff function. In Section II-C, solution for game problem is given. The testing simulation results and analysis are included in Section III. Finally, Section IV concludes the work of this paper.

II. Decision Making Method Design

Fig. 1 shows the typical scenario of unsignalized intersection. WV, NV, EV, SV are the leading straight vehicles from four directions and are all assumed as connected autonomous vehicles. Potential conflict zones exsits between the future trajectories of each two vehicles, which shown as red dot part. The positions of each vehicle are (x_w, y_w), (x_n, y_n), (x_e, y_e), (x_s, y_s) respectively; a_w, a_n, a_e, a_s are the accelerations of each vehicle respectively and l_w, l_n, l_e, l_s are the distances between vehicles to corresponding conflict zone respectively.

The driving mode is adopted as L4 and the related assumptions are adopted as follows: 1) vehicles share the complete driving state (speed, position and so on) and other necessary information such as driving style with each other via V2V communication and each vehicle has the knowledge of the payoff of other vehicles, which is the necessary condition for Stackelberg game [34], [35]; 2) each vehicle follows the pre-planned path (on the basis of origin lane and target lane) to pass through the intersection and adjust the speeds along

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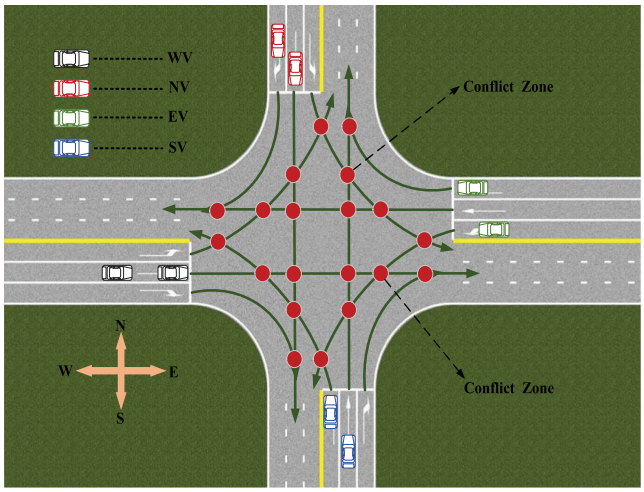

Fig. 1. Typical scenario at the unsignalized intersection.

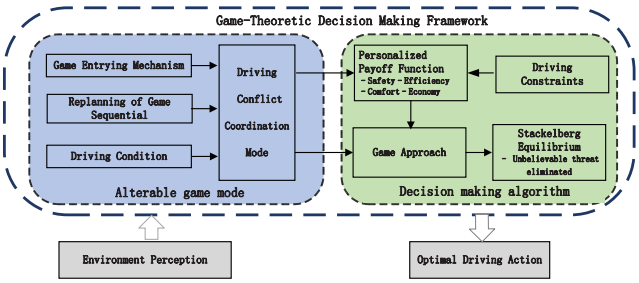

Fig. 2. Decision making framework at the unsignalized intersection.

the path to avoid the conflicts with the other vehicles, which is commonly adopted in the studies [36]; 3) left turn behavior is only allowed for the vehicles entering the left-turn lane at intersection; 4) lane changing behavior is not allowed in the period of driving within the intersection.

Fig. 2 shows the novel decision framework aiming to resolve the potential driving conflicts to realize the safe and orderly passage of connected autonomous vehicles at the unsignalized intersections. The decision making framework mainly consists of three parts: 1) The alterable game mode at unsignalized intersection is established by combining the game entrying mechanism and the real-time adjustment of sequential order. 2) Considering the diversification of driving styles and their influence on decision-making, a game payoff matrix including four indicators integrating driving styles is designed. 3) Solve the Stackelberg equilibrium of all vehicles and exclude the unbelievable threat by backward induction to obtain a set of optimal driving actions (optimal acceleration and speed) of each vehicle.

A. Establishment of Alterable Game Mode

1) Design of Game Entrying Mechanism: The first step of the game-based decision-making algorithm is to establish the game entrying mechanism to determine the autonomous vehicles participating in the game. Previous similar studies simply rely on location division, which is ill-considered. On the one hand, when the distance between the intersection and corresponding vehicle is too large while other vehicles

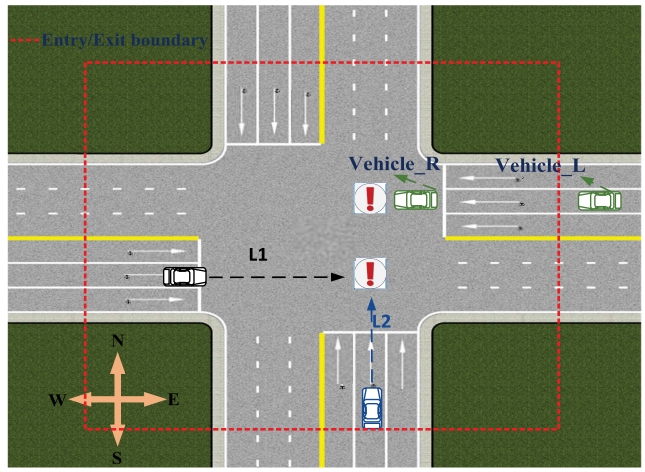

Fig. 3. Schematic diagram of non-game vehicle.

are close to the intersection, the vehicle does not need to participate in the game, which is shown as *Vehicle_L* in Fig. 3. On the other hand, although corresponding vehicle has entered the intersection, if there is no strong game conflict with other vehicles, and there is no need to participate in the game either, which is shown as *Vehicle_R* in Fig. 3. Therefore, considering the particularity of intersection structure and principle of game problem, the game entrying mechanism based on the two-level entry threshold is designed.

On the condition that vehicles are particularly far away from the center of the intersection, there is no need to participate in the game, so first level threshold is the distance threshold which is designed as follow

$$\log_{dis} = \begin{cases} 1, & s_{entry} \leq X_{vehicle,i} \leq s_{exit} \\ 0, & else \end{cases} \tag{1}$$

where $X_{vehicle,i}$ is the current head position of vehicle i, s_{entry} and s_{exit} are the set entry/exit position of intersection respectively.

On the other hand, if the time difference between vehicle and corresponding conflict vehicle arriving at the conflict position is large, participating in the game is needless either, so the relative arrival time (RAT) acts as the second-level threshold and is designed as follow

$$RAT_i = \left| \frac{X_{confl,ij} - X_{vehicle,i}}{V_{vehicle,i}} - \frac{X_{confl,ij} - X_{vehicle,j}}{V_{vehicle,j}} \right| \tag{2}$$

$$\log_{ris} = \begin{cases} 1, & RAT_i \leq RAT_{safe} \\ 0, & else \end{cases} \tag{3}$$

where $X_{confl,ij}$ is the potential conflict position of vehicle i and corresponding vehicle j, $V_{vehicle,i}$ is the current speed of vehicle i, RAT_i is the relative arrival time of conflict vehicles and RAT_{safe} is the corresponding safe threshold.

Game entry threshold is designed according to first-level threshold and second-level threshold as follow

$$\log_{game} = \log_{dis} \& \log_{ris} \tag{4}$$

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TABLE I
COORDINATION MODES IN DIFFERENT INTERSECTION SCENARIOS

Driving scenarios at intersection	Coordination mode
At least three vehicles driving away from conflict area	personal mode for all vehicles
Two vehicles driving away in the opposite direction	personal mode for all vehicles
Two vehicles driving away not in the opposite direction	personal mode for off vehicles; game mode for other vehicles
One or none vehicle driving away from conflict area	game mode for all vehicles

where game entry threshold $\log_{\text{game}} = 1$ only under the condition that the distance threshold $\log_{\text{dis}}$ and risk threshold $\log_{\text{ris}}$ both meet.

2) Construction of Alterable Game Mode: On the basis of game entrying mechanism, the vehicles which entered the intersection but needless to participate in the game is screened out. Further, different game modes based on vehicle driving conditions and specific intersection scenarios are constructed, which is shown as Table I. It is worth noting that the vehicle in the Table I refers to that screened by the game entrying mechanism of equation (4), not all vehicles in the intersection.

In the condition that at least three vehicles driving away from the conflict area, namely off vehicles, the personal mode is selected for all vehicles to meet the personalized needs as there is no need to interact with others. Besides, if there are only opposite vehicles in the intersection, no driving conflict will generate between the corresponding vehicles, so there is no need to enter the game mode either. Once other vehicles driving in or other conditions that will constitute a driving conflict, the game mode is switched to ensure coordinated and safe passage of all vehicles. Through the construction of alterable game mode, the complexity of the game mode is reduced on the premise of ensuring safety.

3) Replanning of Game Sequential Order: After the corresponding vehicles and game mode are determined, the game sequence is further adjusted in every game stage according to the specific driving condition. Specifically, the time to reach the conflict position (TTR) was selected to determine the priority of vehicles as follow

$$TTR_i = \frac{X_{conflict,i} - X_{vehicle,i}}{V_{vehicle,i}\cos(\theta_i)} \tag{5}$$

where $X_{conflict,i}$ is the potential conflict position of vehicle i. $X_{vehicle,i}$ and $V_{vehicle,i}$ are the current head position and speed of vehicle i respectively, θ_i is the angle between vehicle i and the axis. The vehicle with the smaller TTR is considered to be the priority by considering the actual driving scenario, which is more conducive to choose the driving action with greater payoff for urgent vehicle to avoid the occurrence of adverse situations, especially collisions, to ensure the safe passage of vehicles.

B. Design of Payoff Function

Through the game entrying mechanism and alterable game mode mentioned above, participating or non-participating game vehicles are determined. Non-game vehicles are able to drive at any speed within the limitation as there is no conflict relationship, while game-vehicle will try to choose the driving action that makes payoff maximized in every game stage. It is worth pointing out that relevant studies usually ignore the different driving needs of different vehicles while it is fully considered in the design process below, in which four payoff indicators are designed combined with actual driving needs.

1) Design of Payoff Indicator:

a) Safety payoff indicator: Safety is the primary condition to be ensured by drivers in the driving process. In the specific scenario of unsignalized intersection, the larger arrival time lag at the conflicting zone between the corresponding vehicles usually means the safer it is. Denote τ_i as the time for the former vehicle to leave the conflict zone, specifically, the rear of the former vehicle leaving the zone, which is shown as follow

$$\tau_i = \sqrt{\left(\frac{v_i^{n-1}}{a_i^n}\right)^2 + 2\frac{d_i + l_i + W}{a_i^n}} - \frac{v_i^{n-1}}{a_i^n} \tag{6}$$

where a_i^n and v_i^n are the acceleration and speed of vehicle i at stage n respectively, l_i is the length of vehicle, W is the width of lane, d_i is the distance from vehicle i to conflict zone.

Similarly, denote t_i as the time for the latter vehicle to arrive, that is, the time when the front of the latter vehicle arriving at the conflict zone

$$t_i = \sqrt{\left(\frac{v_i^{n-1}}{a_i^n}\right)^2 + 2\frac{d_i}{a_i^n}} - \frac{v_i^{n-1}}{a_i^n} \tag{7}$$

On the premise of ensuring safety, aggressive drivers tend to obtain the right of way and pass the intersection first, while conservative drivers are more willing to surrender the right of way for safety and let conflicting vehicles pass first. Based on the consideration mentioned above, driving safety payoff indicator f_s is designed as follow

$$f_s = \begin{cases} \frac{t_j - \tau_i + \Delta t_{\max}}{2\Delta t_{\max}}, \ if \ t_j \geq \tau_i & driving\ style: aggressive \\ \frac{t_i - \tau_j}{2\Delta t_{\max}}, \quad if \ t_i > \tau_j & \\ \frac{|t_j - \tau_i|}{\Delta t_{\max}} & driving\ style: normal \\ \frac{t_j - \tau_i}{2\Delta t_{\max}}, \quad if \ t_j \geq \tau_i & driving\ style: cautious \\ \frac{t_i - \tau_j + \Delta t_{\max}}{2\Delta t_{\max}}, \ if \ t_i > \tau_j & \end{cases} \tag{8}$$

where Δt_{max} is the max difference of arrival time between conflicting vehicles.

b) Driving efficiency payoff indicator: On the premise of ensuring safety, drivers usually try to complete the driving task with the minimum cost of driving time, that is, to pursue high driving efficiency. At the same time, reference [37] points out that drivers with different driving styles usually have different expected speeds, which means that the higher speed is not always better for different drivers. Therefore, the driving efficiency payoff indicator f_e is designed by considering the

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influence of both two aspects as follow

$$f_e = W_v \cdot \frac{V_i^n}{V_{\max}} + W_b \cdot \left(1 - e^{-\frac{1}{V_i^n - V_{i,best}}}\right) \tag{9}$$

where V_{max} is the limit speed; $V_{i,best}$ is the expected driving speed of vehicle i and it gradually decreases from aggressive style to normal style (less aggressive style and less cautious style respectively) to cautious style; W_v, W_b respectively represent the corresponding weight coefficient.

c) Driving comfort payoff indicator: Frequent or excessive changes of acceleration and deceleration will usually make the driving process unsmooth, thus affect the driving experience. Therefore, the driving comfort payoff indicator f_a is designed as follow

$$f_a = \begin{cases} 1 - \dfrac{\left|a_i^n - a_i^{n-1}\right|}{a_{\max} - a_i^{n-1}}, & \text{if } a_i^{n-1} \le \bar{a} \\ 1 - \dfrac{\left|a_i^n - a_i^{n-1}\right|}{a_i^{n-1} - a_{\min}}, & \text{if } a_i^{n-1} > \bar{a} \end{cases} \tag{10}$$

$$\bar{a} = \frac{a_{\min} + a_{\max}}{2} \tag{11}$$

where a_{max} and a_{min} are the maximum and minimum accelerations of vehicle respectively.

d) Driving economy payoff indicator: In order to reduce driving costs, drivers usually choose the driving strategy of low fuel consumption during driving process. As the acceleration of the vehicle is constant in the time interval T, the average driving speed $\bar{v}$ can be obtained as follow

$$\bar{v} = \frac{v_i^{n-1} + v_i^n}{2} = \frac{v_i^{n-1} + v_i^{n-1} + a_i^n T}{2} = v_i^{n-1} + \frac{T}{2} a_i^n \tag{12}$$

Combining with the average driving speed $\bar{v}$ mentioned above, the fuel consumption rate $F(\bar{v})$ can be obtained as follow [38]

$$F(\bar{v}) = e^{c + d\bar{v} + p\bar{v}^2 + q\bar{v}^3} \tag{13}$$

where c, d, p, q represent the corresponding weight coefficient.

In consideration of the general requirement of low fuel consumption for drivers, the driving economy payoff indicator f_m is designed as follow

$$f_m = 1 - \frac{F(\bar{v}) - F(\bar{v}_{\min})}{F(\bar{v}_{\max}) - F(\bar{v}_{\min})} \tag{14}$$

where $\bar{v}_{\max}$ and $\bar{v}_{\min}$ are the average max speed and min speed of vehicle.

2) Total Payoff Function: The decision making process of the vehicles in choosing the optimal input $u^i = (a^i)$ is assumed to follow the receding horizon strategy. Then, the optimization problem is formulated by using the reward function as follow

$$\mathbb{R}(t) = \sum_{\tau=1}^{N} \xi^{\tau-1} U_i(\tau \,|t) \tag{15}$$

where $U_i(\tau \,|t)$ is the stage reward prediction of time step $t + \tau$ at time t for corresponding vehicle i, N is the the prediction horizon, ξ is the discounting factor and $\xi \in [0, 1]$. In particular, on the basis of four payoff indicators mentioned above, $U(\tau \,|t)$ is designed as a linear combination of four payoff indicators as follow

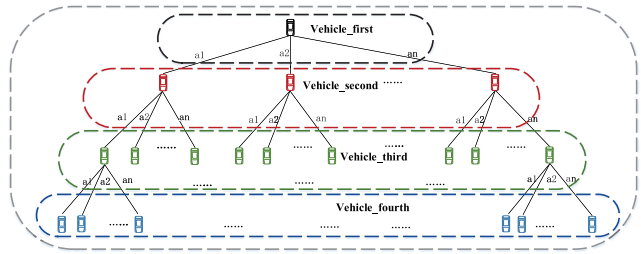

Fig. 4. Schematic diagram of game process.

$$U_i(\tau \,|t) = C_s \cdot f_{s,i}(\tau \,|t) + C_e \cdot f_{e,i}(\tau \,|t) + C_a \cdot f_{a,i}(\tau \,|t) + C_m \cdot f_{m,i}(\tau \,|t) \tag{16}$$

where C_s, C_e, C_a, C_m are the weight coefficients of four corresponding payoff indicators and $C_s + C_e + C_a + C_m = 1$. It's worth noting that the weight varies according to the driver's style, reflecting the different driving needs. Typically, normal drivers tend to balance the payoff of all four indicators, while aggressive drivers tend to pay more attention to driving efficiency and ignore fuel consumption costs to some extent.

C. Solution for Game Problem

Different from the static game in which players take actions simultaneously, the dynamic game has a sequential relationship, and the game process expansion diagram is shown as Fig. 4. Therefore, the payoff of game players is usually affected by other players in dynamic game. Specifically, players with priority action right usually has the advantage in that they are able to directly take the action s_1^* that with the greatest payoff, on the basis of predicting the actions of the later players

$$s_1^* = \operatorname{argmax}\left(\max u_1\left(s_1, S_2^1(s_1), S_3^1(s_1) \ldots S_N^1(s_1)\right)\right) \tag{17}$$

$$u_1\left(s_1^*, S_2^1(s_1^*), S_3^1(s_1^*) \ldots S_N^1(s_1^*)\right) \ge u_1\left(s_1, S_2^1(s_1), S_3^1(s_1) \ldots S_N^1(s_1)\right), \forall s_1 \in S_1 \tag{18}$$

where s_i is the action take by player i, S_i is the action set of player i, u_i is the payoff function of player i, $S_j^i(s_i)$ is the action set of player j in the condition that former player takes the action i. Other players have to take the driving action (s_2^*, s_3^* for example) that with the greatest payoff, in the condition that specific action that has been selected by former player and predicting the action that later player will take

$$u_2\left(s_1^*, s_2^*, S_3^2(s_1^*, s_2^*) \ldots S_N^2(s_1^*, s_2^*)\right) \ge u_2\left(s_1^*, s_2, S_3^2(s_1^*, s_2) \ldots S_N^2(s_1^*, s_2)\right), \quad \forall s_2 \in S_2 \tag{19}$$

$$u_3\left(s_1^*, s_2^*, s_3^* \ldots S_N^3(s_1^*, s_2^*, s_3^*)\right) \ge u_3\left(s_1^*, s_2^*, s_3 \ldots S_N^3(s_1^*, s_2^*, s_3)\right)\Big\}, \quad \forall s_3 \in S_3 \tag{20}$$

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The last player tend to take the action s_n^* with the greatest payoff after all the players have took the actions successively

$$u_N\left(s_1^*, s_2^*, s_3^* \ldots \quad s_{N-1}^*, s_N^*\right) \geq u_N\left(s_1^*, s_2^*, s_3^* \ldots s_{N-1}^*, s_N\right)\Big\}, \quad \forall s_N \in S_N \tag{21}$$

Furthermore, if there exists a combination of control actions $\left(s_1^*, \ldots, s_{i-1}^*, s_i^*, s_{i+1}^*, \ldots, s_n^*\right)$ in the non-cooperative game in which any player is not able to obtain more payoff u_i by changing the control action unilaterally, this combination is defined as Nash equilibrium

$$u_i\left(s_1^*, \cdots, s_{i-1}^*, s_i^*, s_{i+1}^*, \cdots, s_n^*\right) \geq u_i\left(s_1^*, \cdots, s_{i-1}^*, s_p^*, s_{i+1}^*, \cdots, s_n^*\right) \tag{22}$$

Specially, for the Stackelberg game problem, assuming S^{leader} is the action set of leader, $\Psi(s_f)$ is the optimal action set of follower for $\forall s_l \in S^{leader}$, corresponding to the maximum payoff $U^{follower}\left(s_l, s_f\right)$. The equilibrium $\left(s_l^*, s_f^*\right)$ is attained only when the following condition met

$$U^{follower}\left(s_l^*, s_f^*\right) \geq U^{follower}\left(s_l, s_f\right) \quad \begin{cases} s_f^* \in \Psi(s_f) \\ \forall s_l \in S^{leader}, \quad \forall s_f \in S^{follower} \end{cases} \tag{23}$$

However, in the dynamic game process, there usually exists unbelievable threat, which refers to some equilibriums that are impossible to be reached in Nash equilibrium as all the players are assumed as rational. The principle of backward induction is that the analysis starts from the player in the last stage and backwards to the first player step by step, which is considered to be the most effective method to eliminate the unbelievable threat. Therefore, the decision making process for safe and orderly passage of autonomous vehicles at unsignalized intersection is summarized as Algorithm 1.

Algorithm 1 Decision Making of Autonomous Vehicles for Safe Passage at the Unsignalized Intersection

Input: Current position X_i of all vehicles;
Speed and acceleration V_i, a_i of all vehicles.
Output: Decision results u_j^* for all vehicles.

```
for i = 1 : m do
    Determine the game-vehicle H_game and non-game vehicle H_personal;
    Determine the coordination mode Γ;
    Adjust the game sequential order;
    for j = 1 : n do
        Calculate the payoff indicator f_{s,j}, f_{e,j}, f_{a,j}, f_{m,j};
        Calculate the total payoff U_j at current action set S_j;
    Calculate the Stackelberg equilibrium u_j*;
    Eliminate the unbelievable threat among equilibriums;
```

III. Simulation Results and Discussion

Prescan has the advantage of facilitating the construction of various simulation scenarios, and has many built-in vehicle dynamics models, which can perform well in the construction of automatic driving simulation scenarios and control object models. Matlab has been widely used to write and debug various control strategies for its powerful computing power. Therefore, the co-simulation platform of Matlab and Prescan is established to verify the effectiveness of the proposed method for the adaptability to simulate the individual behavior of autonomous vehicles and build up the scenario of unsignalized intersection.

The working flow of the co-simulation platform is set as follows: Prescan first sends the state information of the vehicles to the Simulink decision making model, and the Simulink decision making model calculates the corresponding control signal by the received information. Then the states of the vehicles in Prescan and the positions in the simulated environment animation are updated by the control signal received from Simulink decision making model through the virtual CAN bus and the process repeats until the end of the simulation.

TABLE II
Main Parameters

Parameter	Value
$a_{\max}$	3 m^2
$a_{\min}$	-3 m^2
w	4 m
l	5 m
T	0.3 s
$v_{\max}$	25 m/s

To verify the effectiveness of the proposed decision making method of autonomous vehicles at unsignalized intersections, a series of driving scenarios including five cases are carried out under the co-simulation environment of Matlab and Prescan. Some main parameters are listed in Table II and the payoff weight coefficients C_s, C_e, C_a, C_m are set as 0.4, 0.3, 0.15, 0.15 for aggressive style, 0.4, 0.2, 0.2, 0.2 for normal style and 0.4, 0.1, 0.25, 0.25 for cautious style respectively.

A. Case 1: Differentiated Driving Style Scenario

In this case, the differentiated driving style of autonomous vehicles and its influence on decision results are discussed. Specifically, the distance from each vehicle to the center of the intersection are both 125m. The initial speed and acceleration are both 12m/s and 0m/s^2 respectively. The driving style of WV, EV, SV, NV are set as aggressive, less aggressive, less cautious, cautious respectively.

Fig. 5-7 show the acceleration, speed and payoff of each vehicle respectively, which demonstrate that WV selects the maximum acceleration at the beginning stage to get the right of way as the most aggressive among all vehicles, and the acceleration gradually decreases to 0 as approaches the desired

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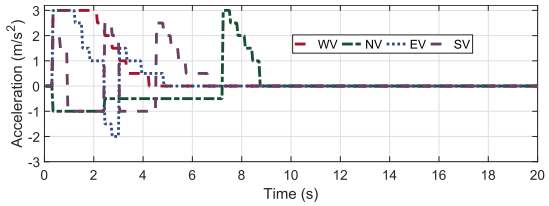

Fig. 5. Accelerations of autonomous vehicles in Case 1.

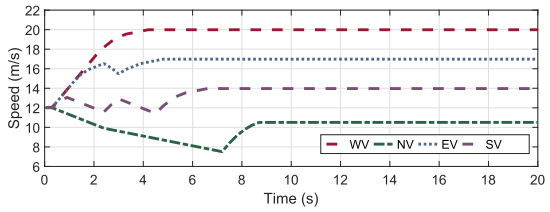

Fig. 6. Speeds of autonomous vehicles in Case 1.

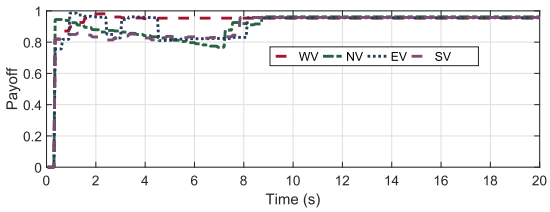

Fig. 7. Payoff of autonomous vehicles in Case 1.

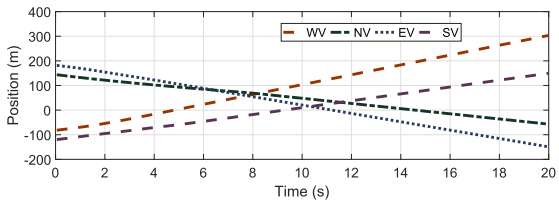

Fig. 8. Positions of autonomous vehicles in Case 1.

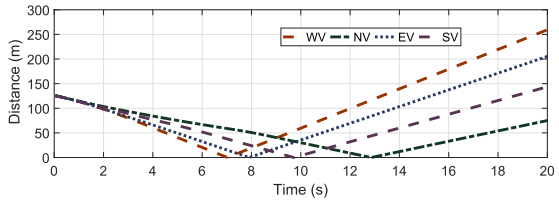

Fig. 9. Distances of autonomous vehicles in Case 1.

speed. The speed of EV and SV varies relatively larger as it focuses on all driving indicators at the same time, so the total payoff changes more smoothly either. NV tend to slow down to give up the right of way due to its cautious style and -1m/s^2 is selected, and when other vehicles are about to pass through the intersection, it selects the maximum acceleration 3m/s^2 to start accelerating till reaching the desired speed to pass through the intersection. It is also noted that although NV give up the right of the way due to cautious style, it also has lower fuel consumption than the aggressive vehicle WV, 16.35mL and 21.02mL respectively.

Fig. 8-9 are simulation curves of position and distance from corresponding conflict zone of each vehicle. It can be seen

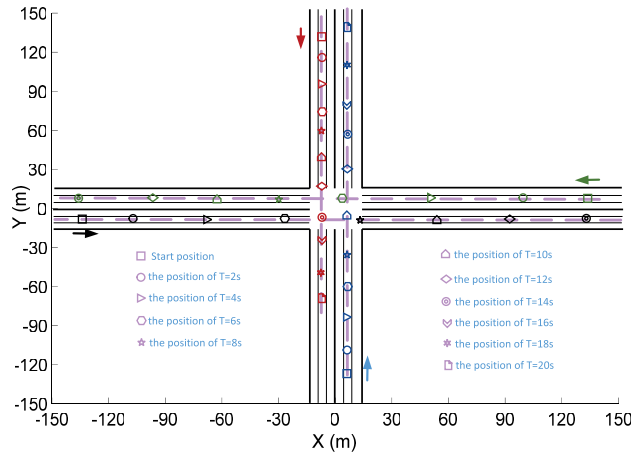

Fig. 10. Trajectory projection of autonomous vehicles in Case 1.

TABLE III
STATISTICAL RESULTS IN CASE 1

Vehicle	Number of interaction	Exiting speed (m/s)	Passing time (s)
W_V	23	19.99	7
N_V	32	10.51	12.9
E_V	26	16.98	7.9
S_V	32	13.97	9.7

that WV, EV, SV, NV pass through the intersection conflict area in sequence. When WV reaches the conflict zone, the distance of conflicting vehicle NV and SV are 60m and 38m respectively; When NV reaches the conflict zone, the distance of conflicting vehicle WV and EV are 115.5m and 83m respectively; When EV reaches the conflict zone, the distance of conflicting vehicle NV and SV are 51.1m and 24.7m respectively; When SV reaches the conflict zone, the distance of conflicting vehicle WV and EV are 54.5m and 33m respectively. It can be seen that all distances are larger than safe distance, so all vehicles are able to pass the intersection without collision.

Fig. 10 represents the motion trajectory projection of each vehicle. It can be seen that WV and EV gradually accelerate and get the right of way in priority. SV changes the speed constantly by taking all relevant indicators into consideration and NV concedes the right of way directly, slows down and waits for other vehicles to pass.

Table III shows the exiting speed, passing time and total number of game interaction of each vehicle in scenario 1, which demonstrates that the different driving style has a direct impact on the game decision results. On the basis of same initial condition, more aggressive vehicle will has more priority after several game stages, so it tends to choose to accelerate and get the right of way to ensure its priority through the intersection; in contrast, the cautious vehicle tends to restricted by other vehicles and give up the right of way to ensure driving safety, which is consistent with the actual driving decision.

B. Case 2: Differentiated Driving Status Scenario

The differentiated driving status of autonomous vehicles and its influence on decision results are discussed in this case.

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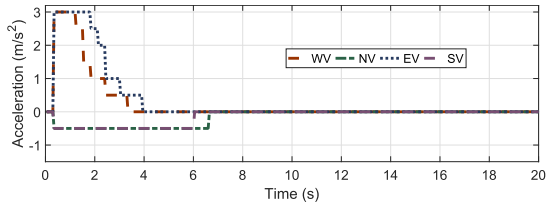

Fig. 11. Accelerations of autonomous vehicles in Case 2.

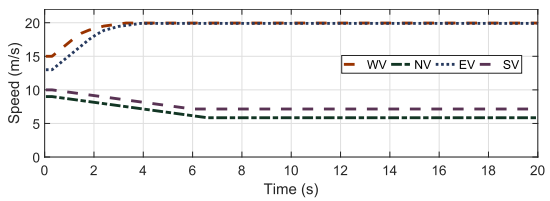

Fig. 12. Speeds of autonomous vehicles in Case 2.

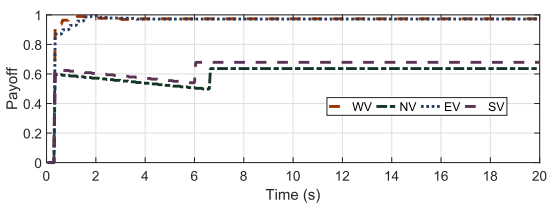

Fig. 13. Payoff of autonomous vehicles in Case 2.

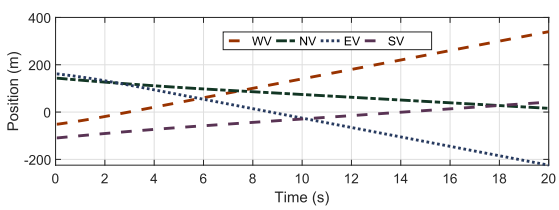

Fig. 14. Positions of autonomous vehicles in Case 2.

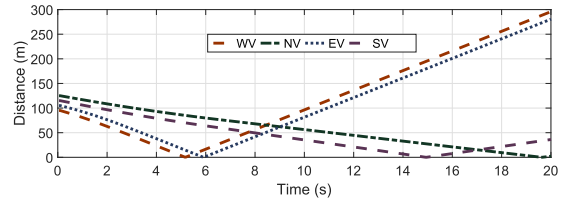

Fig. 15. Distances of autonomous vehicles in Case 2.

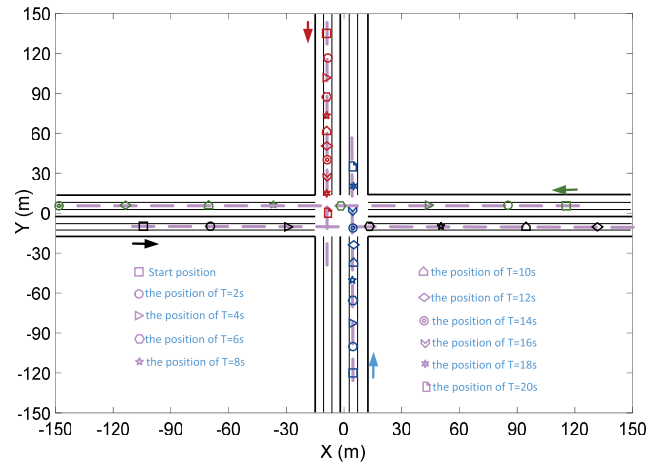

Fig. 16. Trajectory projection of autonomous vehicles in Case 2.

TABLE IV
STATISTICAL RESULTS IN CASE 2

Vehicle	Number of interaction	Exiting speed (m/s)	Passing time (s)
W_V	17	19.95	5.2
N_V	50	5.85	19.65
E_V	20	19.9	5.95
S_V	50	7.15	15

Specifically, the driving style of all vehicles are both set as aggressive. The initial speed of WV, NV, EV, SV are set as 15m/s, 9m/s, 11m/s, 10m/s respectively. The distance of WV, NV, EV, SV to the center of the intersection are set as 95m, 125m, 105m, 115m respectively.

Fig. 11-13 show the acceleration, speed and payoff of each vehicle respectively. It can be seen that as WV and EV have both distance advantage and speed advantage for initial state, so they are able to have the advantage to choose the maximum acceleration to reach the desired speed of 20m/s and get the right of way to pass the intersection first and exit the game. However, due to long initial distance and low speed, NV and SV have the latter sequential order in the dynamic game. NV and SV can only choose -0.5m/s^2 to reduce speed to yield the right of way, which is obviously not optimal individually and has a relative low profit. Fig. 14-15 are simulation curves of position and distance from corresponding conflict zone of each vehicle. It can be seen that WV and EV pass through the intersection first, NV and SV pass through the intersection later, which is line with the initial state condition. The passing time to corresponding conflict zone for WV, EV, SV, NV are 5.2s, 6s, 14.9s and 19.6s respectively. The smallest time difference among the conflicting vehicles is 8.9s, which is much larger than the safety requirement, so the safe passage of all vehicles are ensured.

Fig. 16 represents the motion trajectory projection of each vehicle, which shows that WV and EV gradually speed up to pass through the intersection, while NV and SV slow down and finally pass through the intersection at a constant speed. Table IV is the corresponding statistical results of each vehicle in scenario 2, which shows that WV and EV pass the intersection with a relative higher speed and shorter time than NV and SV. The decision results are consistent with actual driving scenario that vehicles with more advantages in the initial state tend to has the priority to choose the driving action to speed up and get the right of way, while other vehicles usually give up the right of way and then pass through the intersection to avoid the potential conflict.

C. Case 3: Differentiated Driving Status & Driving Style Scenario

In this case, the most complex driving scenario is discussed, in which driving status and driving style of autonomous vehicles are all differentiated. Specifically, the driving style of WV, EV, SV, NV are set as aggressive, less aggressive, less

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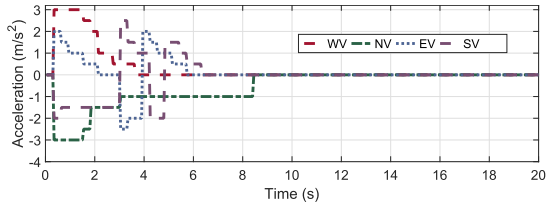

Fig. 17. Accelerations of autonomous vehicles in Case 3.

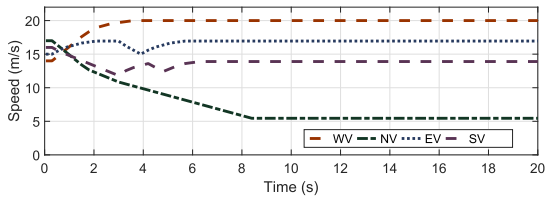

Fig. 18. Speeds of autonomous vehicles in Case 3.

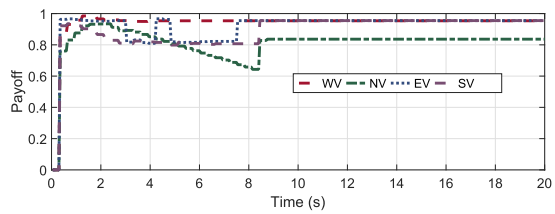

Fig. 19. Payoff of autonomous vehicles in Case 3.

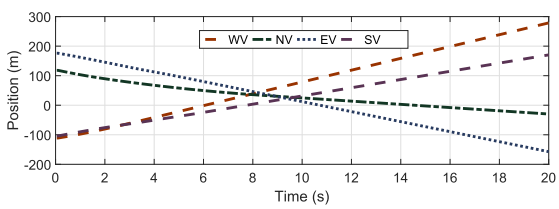

Fig. 20. Positions of autonomous vehicles in Case 3.

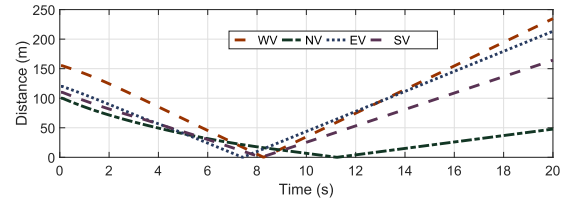

Fig. 21. Distances of autonomous vehicles in Case 3.

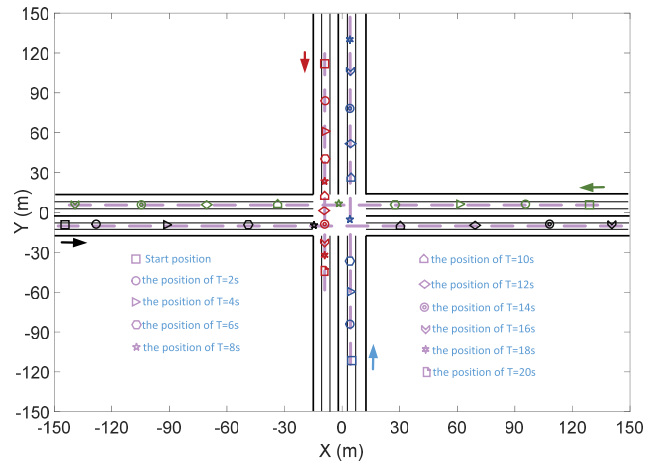

Fig. 22. Trajectory projection of autonomous vehicles in Case 3.

TABLE V
STATISTICAL RESULTS IN CASE 3

Vehicle	Number of interaction	Exiting speed (m/s)	Passing time (s)
W_V	28	20	8.25
N_V	28	5.45	11.25
E_V	25	16.95	7.45
S_V	27	13.9	8.15

cautious, cautious respectively. The initial speed of WV, NV, EV, SV are set as 14m/s, 17m/s, 15m/s, 16m/s respectively. The distance of WV, NV, EV, SV to the center of the intersection are set as 155m, 100m, 120m, 110m respectively.

Fig. 17-19 show the acceleration, speed and payoff of each vehicle respectively. It can be seen that though WV does not have an advantage in its initial state at the beginning stage of the game for long distance and low speed compared with NV, it is still able to choose the maximum acceleration strategy of 3m/s^2 as its aggressive driving style and accelerate to the desired speed of 20m/s to pass through the intersection at a constant speed. While NV chooses the deceleration strategy to give up the right of way until WV passes the intersection and exits the game at 8.4s. The speed curve shown in Fig. 18 also demonstrates that the autonomous vehicles are able to adjust the speeds with the change of surrounding driving senario during the driving process, and the speeds vary relatively flat and range of variation is reasonable.

Fig. 20-21 are simulation curves of position and distance from corresponding conflict zone of each vehicle, which shows that the distance between conflicting vehicles are 14.4m, 22.7m, 28.3m and 52.9m respectively, both larger than the safe distance, ensuring all vehicles pass the intersection safely and without collision.

Fig. 22 represents the motion trajectory projection of each vehicle, which shows that WV, EV, SV have the similar trend of speed, while NV slows down due to cautious style and passes through the intersection last. Table V shows the corresponding statistical results of each vehicle in scenario 3, which shows that although WV and EV have a disadvantage in the initial state, they are able to pass the intersection with higher speed (20m/s, 16.95m/s respectively) and shorter passing time (8.25s, 7.45s respectively) than NV (5.45m/s, 11.25s respectively) and SV (13.9m/s, 8.15s respectively), which is consistent with the actual driving condition that relatively aggressive driving style is able to make up for the disadvantage of the initial state to some degree, and then make it still able to get right of the way and pass the intersection first.

D. Case 4: Mixed Driving Status & Driving Style Scenario

As the complex influence of driving status and driving style on decision results and in order to avoid the limitation of single scenario, another different mixed scenario is set and discussed in this case. Specifically, the driving style of

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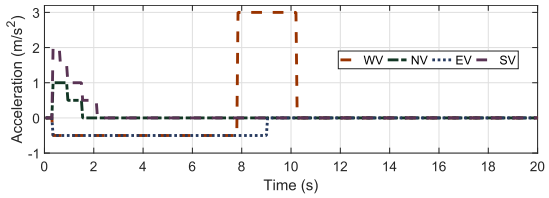

Fig. 23. Accelerations of autonomous vehicles in Case 4.

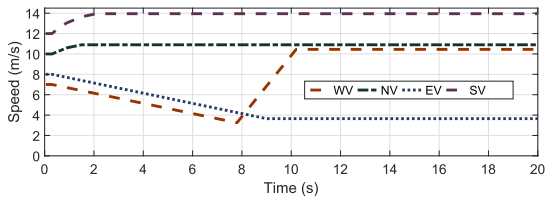

Fig. 24. Speeds of autonomous vehicles in Case 4.

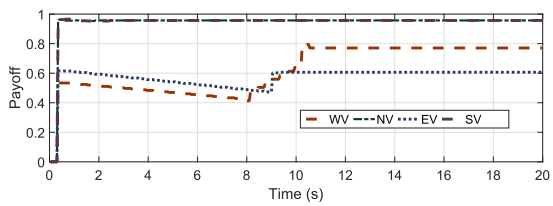

Fig. 25. Payoff of autonomous vehicles in Case 4.

WV, EV, SV, NV are set as aggressive, less aggressive, less cautious, cautious respectively. The initial speed of WV, NV, EV, SV are set as 7m/s, 10m/s, 8m/s, 12m/s respectively. The distance to corresponding conflict zone of WV, NV, EV, SV are set as 155m, 95m, 115m, 105m respectively.

Fig. 23-25 show the acceleration, speed and payoff of each vehicle respectively. For NV and WV, it can be seen that NV chooses the acceleration strategy of 1m/s^2 to maximize the payoff as it has the priority to make decisions in the game for the advantage in initial state. While WV is not able to choose the optimal action for its own payoff as in the latter sequential order, so it has to slow down at -0.5m/s^2 to give up the right of way and finally pass the intersection at the speed of 10.5m/s. It is similar for SV and EV. SV is able to choose the 2m/s^2 and gradually accelerates to the desired speed of 14m/s for the advantage in initial state, while EV has to choose the -0.5m/s^2 and gives up right of the way at the same time.

Fig. 26-27 are simulation curves of position and distance from respective conflict zone of each vehicle, which shows that SV, NV and WV pass through the intersection at 7.8s, 8.9s and 19.7s respectively. Fig. 28 represents the motion trajectory projection of each vehicle, which shows that SV, NV pass through the intersection in an accelerating trend first and all vehicles pass the intersection in a safe and orderly way. Table VI shows the corresponding statistical results of each vehicle in scenario 4. It can be seen that although the driving style of SV and NV is less aggressive, they are able to get the right of the way and pass the intersection first, which is consistent with actual decision scenario that vehicles with a larger initial disadvantage will have to give up right of the way and slow down to pass the intersection later.

TABLE VI
STATISTICAL RESULTS IN CASE 4

Vehicle	Number of interaction	Exiting speed (m/s)	Passing time (s)
W_V	66	10.45	19.65
N_V	30	10.9	8.9
E_V	66	3.65	26.03
S_V	26	13.95	7.75

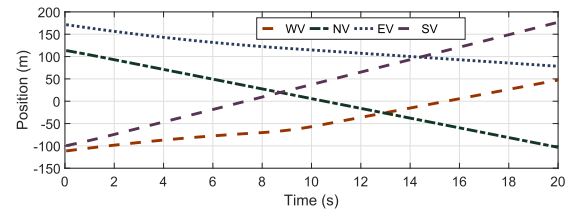

Fig. 26. Positions of autonomous vehicles in Case 4.

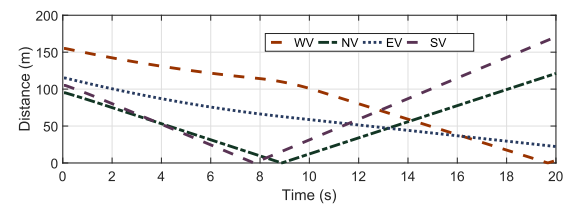

Fig. 27. Distances of autonomous vehicles in Case 4.

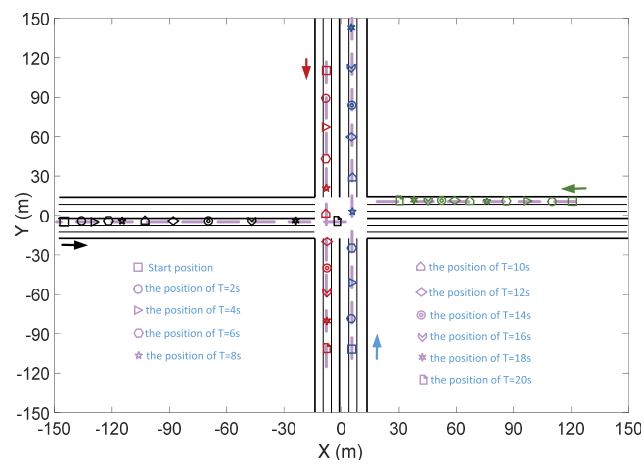

Fig. 28. Trajectory projection of autonomous vehicles in Case 4.

E. Case5: Variational Driving Style Scenario

In order to analyse the impact of driving style on decision results more intuitively, variation of driving style on designated vehicle in same scenario is set and discussed. Specifically, the initial speed and distance to corresponding conflict zone of WV, NV, EV, SV are all set as 15m/s and 125m respectively. The driving style of WV, NV, EV are set as less aggressive and the driving style of SV is set as aggressive, less aggressive and cautious respectively.

Fig. 29-31 show the acceleration, speed and payoff of SV under different driving style respectively. It can be seen that SV choose the maximum acceleration strategy of 3m/s^2 at the first stage, and get the right of way under the aggressive style directly. When the driving style changes to less aggressive, it can be seen that 3m/s^2 is selected at about 1.2s,

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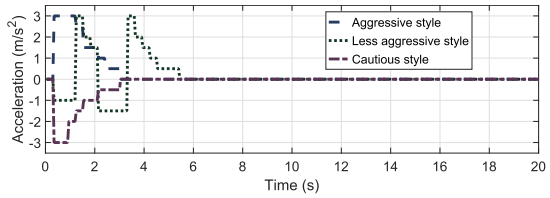

Fig. 29. Accelerations of autonomous vehicles in Case 5.

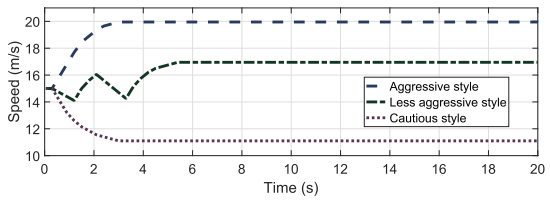

Fig. 30. Speeds of autonomous vehicles in Case 5.

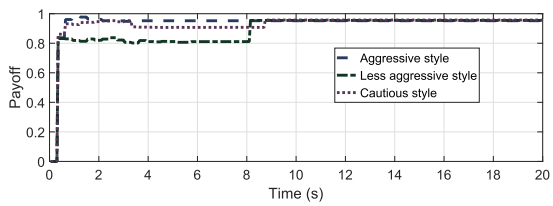

Fig. 31. Payoff of autonomous vehicles in Case 5.

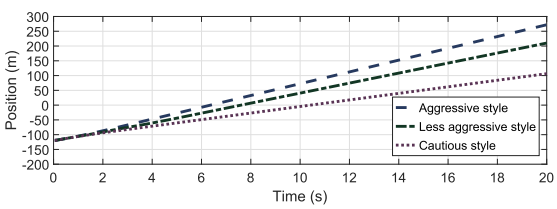

Fig. 32. Positions of autonomous vehicles in Case 5.

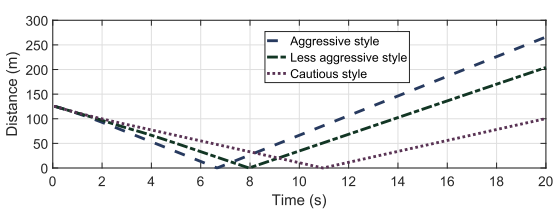

Fig. 33. Distances of autonomous vehicles in Case 5.

and -1.5 m/s^2 is selected during 2.2s-3.25s, which for that all payoff indicators are both considered equally, leading to repeated acceleration and deceleration in the whole strategy selection process. Finally, when the desired speed of 17m/s is reached, the acceleration gradually drops to 0, and keeps a constant speed and approaches the intersection at the current speed. As for condition under cautious driving style, -3m/s^2 is selected to slow down to the desired speed of 11m/s and give up right of the way and passes the intersection latter.

Fig. 32-33 are simulation curves of position and distance from corresponding conflict zone under different driving styles. It can be seen that as the driving style becomes more aggressive, the time to passing the intersection becomes shorter. Table VII is the corresponding statistical results of

TABLE VII
STATISTICAL RESULTS IN CASE 5

Driving style	Number of interaction	Exiting speed (m/s)	Passing time (s)
Aggressive	22	20	6.7
Less aggressive	26	17	7.8
Cautious	36	11	10.9

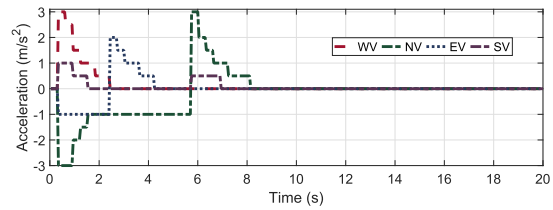

Fig. 34. Accelerations of autonomous vehicles in Case 6.

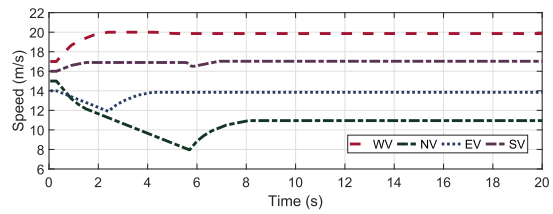

Fig. 35. Speeds of autonomous vehicles in Case 6.

SV under different driving styles in scenario 5. It can be seen that the passing time of SV under three driving styles is 6.65s, 8s and 10.95s respectively. This is consistent with the human decision that the aggressive vehicle tend to choose to accelerate decisively and get the right of way, the normal vehicle usually takes all kinds of factors considered at the same time, while the cautious vehicle tends to give up the right of way, slow down and pass the intersection later. It can also be noted that although the driving status and driving style vary greatly in different cases, the autonomous vehicles are still able to pass the intersection without collision and in a orderly way under the proposed method and driving style and driving status jointly affect and determine the ownership of right of way.

F. Case6: Mixed Straight Vehicle & Turning Vehicle Scenario

In order to further complicate the driving scenario, vehicles going straight and vehicles turning at intersection are taken into consideration at the same time in this case. Specifically, NV and EV take the action of going straight to pass through the intersection, WV and SV take the action of left turn and right turn to pass through the intersection respectively. The driving style of WV, SV, EV, NV are set as aggressive, less aggressive, less cautious, cautious respectively. The initial speed of WV, NV, EV, SV are set as 17m/s, 15m/s, 14m/s, 16m/s respectively. The distance to corresponding conflict zone of WV, NV, EV, SV are set as 85m, 100m, 105m, 95m respectively.

Fig. 34-36 show the acceleration, speed and payoff of each vehicle respectively. It can be seen that as WV and SV have an advantage in the initial state at the beginning stage of the game for short distance and high speed, they are able to choose the acceleration strategy of 3m/s^2 and 1m/s^2 respectively and

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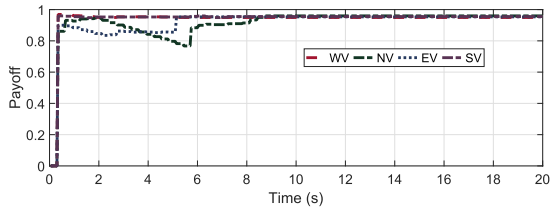

Fig. 36. Payoff of autonomous vehicles in Case 6.

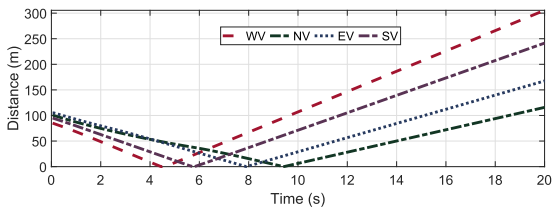

Fig. 37. Distances of autonomous vehicles in Case 6.

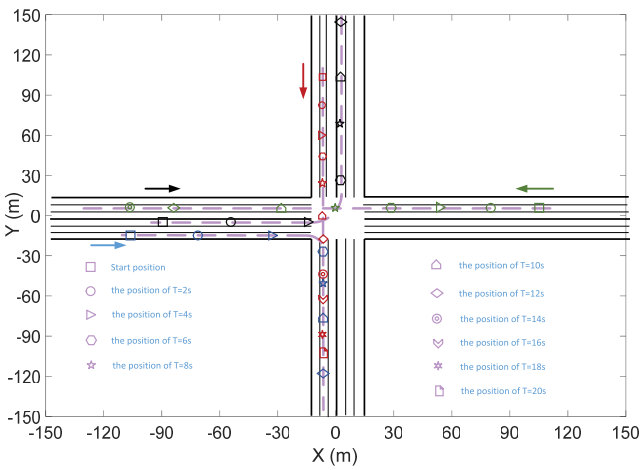

Fig. 38. Trajectory projection of autonomous vehicles in Case 6.

accelerate to the desired speed to pass through the intersection at a constant speed. While NV and EV have to slow down and give up the right of way as in the latter sequential order and cautious style and finally pass the intersection subsequently at the speed of 10.9m/s and 13.8m/s respectively.

Fig. 37 is the simulation curve of distance from corresponding conflict zone of each vehicle. It can be seen that WV, SV, EV, NV pass through the intersection conflict area in sequence. When WV reaches the conflict zone, the distance of conflicting vehicle NV and EV are 49m and 47.2m respectively; When SV reaches the conflict zone, the distance of conflicting vehicle NV is 37.8m; When EV reaches the conflict zone, the distance of conflicting vehicle WV and NV are 65.2m and 16.7m respectively; When NV reaches the conflict zone, the distance of conflicting vehicle EV, WV, SV are 20.7m, 95m and 60.9m respectively. It can be seen that all distances are larger than safe distance, so the safe passage of all vehicles are ensured.

Fig. 38 represents the motion trajectory projection of each vehicle, which shows that WV, SV pass through the intersection in an accelerating trend first and all vehicles pass the intersection without collision and in a orderly way. In order to further verify the effectiveness of the proposed method, the rule-based control method proposed in [32] is adopted as the comparison scheme. The main control rules of the method can be interpreted as follows: If there is no other vehicle in conflict with the autonomous vehicles, the autonomous vehicles will pursue the higher driving speed along the reference path. If there are some vehicles existing the driving conflict with it, the autonomous ego vehicle will give way by maximizing driving distances. Specifically, the acceleration selected by the autonomous vehicles at each time instant is introduced by Algorithm 2 in [32] and is not expanded here. Table VIII and Table IX are the corresponding statistical results of each vehicle under two strategies in scenario 6, which show that although the vehicles that arrive first can pass through in a short time under the rule based control method, it is difficult to avoid the relatively long waiting time of other vehicles, resulting in a longer overall passing time. However, all vehicles under the proposed method can adjust their speed properly and pass the intersection in a orderly way and 49.4% (from 3.7743s to 1.9093s) of the average travel delay is reduced, which shows that the proposed method can not only ensure the safe passage of autonomous vehicles at the unsignalized intersection, but also further reduce the travel delay and improve the traffic efficiency.

TABLE VIII
STATISTICAL RESULTS IN CASE 6

Vehicle	Number of interaction	Exiting speed (m/s)	Passing time (s)
W_V	15	19.86	4.5
N_V	26	10.94	9.4
E_V	26	13.83	7.9
S_V	19	17.02	5.8

TABLE IX
STATISTICAL RESULTS UNDER RULE-BASED CONTROL IN CASE 6

Vehicle	Number of interaction	Exiting speed (m/s)	Passing time (s)
W_V	36	12.35	10.9
N_V	23	12.75	6.9
E_V	22	11.92	6.6
S_V	36	9.86	10.8

IV. CONCLUSION

This paper investigates the conflict resolution problem of autonomous vehicles at unsignalized intersections. A game-theoretic decision making method considering personalized driving preferences considered is designed. On the basis of the designed game entrying mechanism and replanning of game sequential order, the alterable game mode is constructed, which reduce the complexity and improve the effectiveness. Besides, personalized payoff function is designed based on four payoff indicators with the general driving requirements and personalized driving preferences both considered. Six different typical scenarios at the unsignalized intersection were performed under co-simulation environment of Matlab and Prescan. Simulation results show that the proposed method is able to resolve the driving conflicts of autonomous vehicles at the unsignalized intersection with personalized driving preferences met.

Future work will focus on applying the proposed game-theoretic decision making method to other multi-agent

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interaction scenarios to further extend the applied range, such as highway merging, lane-changing and so on.

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Xinghao Lu received the B.E. degree from the College of Communication Engineering, Jilin University, in 2019, where he is currently pursuing the Ph.D. degree with the Department of Control Science and Engineering. His research interests include model predictive control and chassis integrated control of electric vehicle.

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Haiyan Zhao received the B.S. degree in automation and the M.S. and Ph.D. degrees in control theory and control engineering from Jilin University, Changchun, China, in 1998, 2004, and 2007, respectively. She is currently a Professor with Jilin University. Her current research interests include model predictive control, vehicle stability control, and the state estimation of electric vehicles.

Cheng Li received the B.S. degree in automation from Jilin University, Changchun, China, in 2018, where he is currently pursuing the M.S. degree with the Department of Control Engineering. His current research focuses on the conflict coordination of autonomous vehicles.

Bingzhao Gao (Member, IEEE) received the B.S. and M.S. degrees in vehicle engineering from Jilin University, Changchun, China, in 1998 and 2002, respectively, and the dual Ph.D. degree in mechanical engineering from Yokohama National University, Yokohama, Japan, and in control engineering from Jilin University, in 2009. He is currently a Professor with Tongji University. His current research interests include vehicle powertrain control and vehicle stability control.

Hong Chen (Fellow, IEEE) received the B.S. and M.S. degrees in process control from Zhejiang University, Zhejiang, China, in 1983 and 1986, respectively, and the Ph.D. degree in system dynamics and control engineering from the University of Stuttgart, Stuttgart, Germany, in 1997. From 1999 to 2021, she has been a Professor with Jilin University, Changchun, China, where she worked as a Tang Aoqing Professor and the Director of the State Key Laboratory of Automotive Simulation and Control. She is currently a Professor with Tongji University. Her current research interests include model predictive control, optimal and robust control, and nonlinear control and applications in mechatronic systems focusing on automotive systems.

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